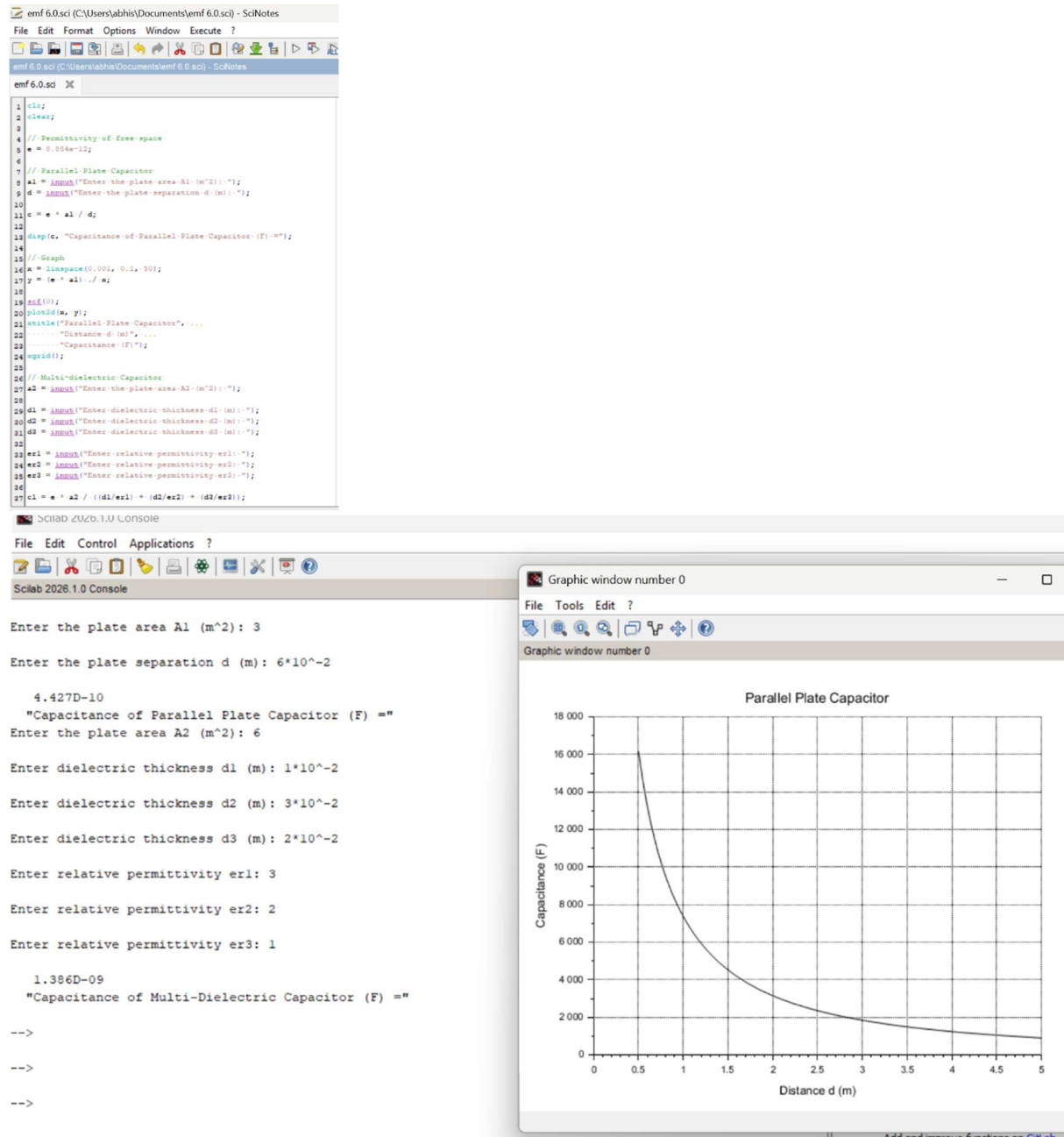


Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant.

Aim: To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.



Formula

$$C = \frac{\epsilon_0 A}{d}$$

Calculation

$$C = \frac{(8.854 \times 10^{-12}) (3)}{0.06}$$

$$= \frac{2.6562 \times 10^{-11}}{0.06}$$

$$C = 4.427 \times 10^{-10} \text{ F}$$

Formula

$$C = \frac{\epsilon_0 A}{\left(\frac{d_1}{\epsilon_{r1}} + \frac{d_2}{\epsilon_{r2}} + \frac{d_3}{\epsilon_{r3}} \right)}$$

$$\frac{d_1}{\epsilon_{r1}} = \frac{0.01}{3} = 0.00333$$

$$\frac{d_2}{\epsilon_{r2}} = \frac{0.03}{2} = 0.015$$

$$\frac{d_3}{\epsilon_{r3}} = \frac{0.02}{1} = 0.02$$

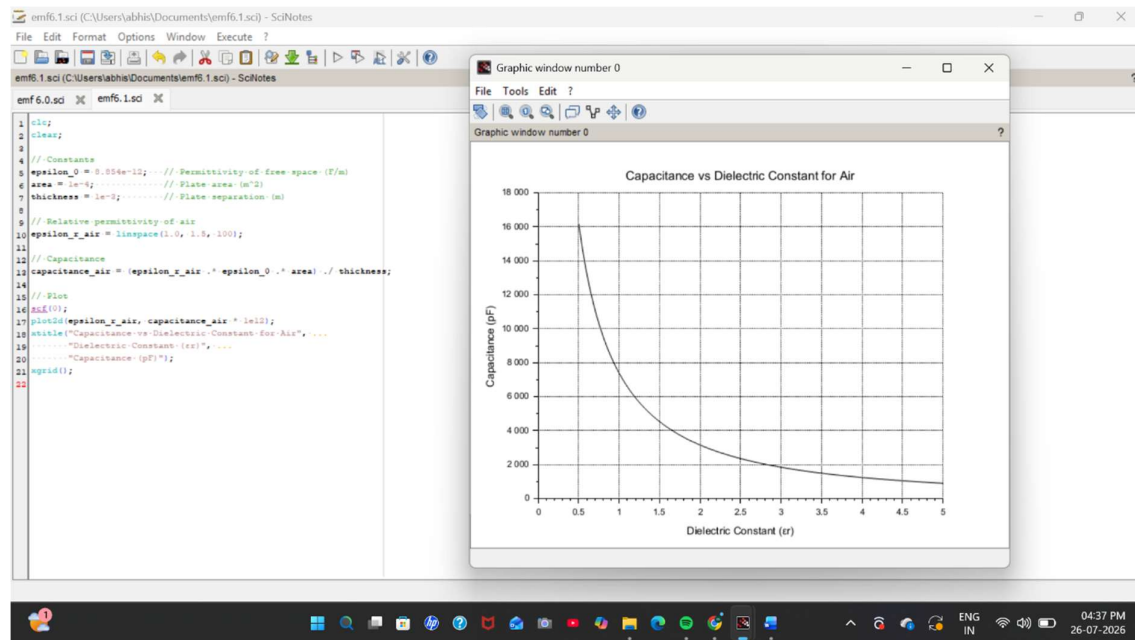
$$0.00333 + 0.015 + 0.02 = 0.03833$$

$$C = \frac{(8.854 \times 10^{-12}) (6)}{0.03833}$$

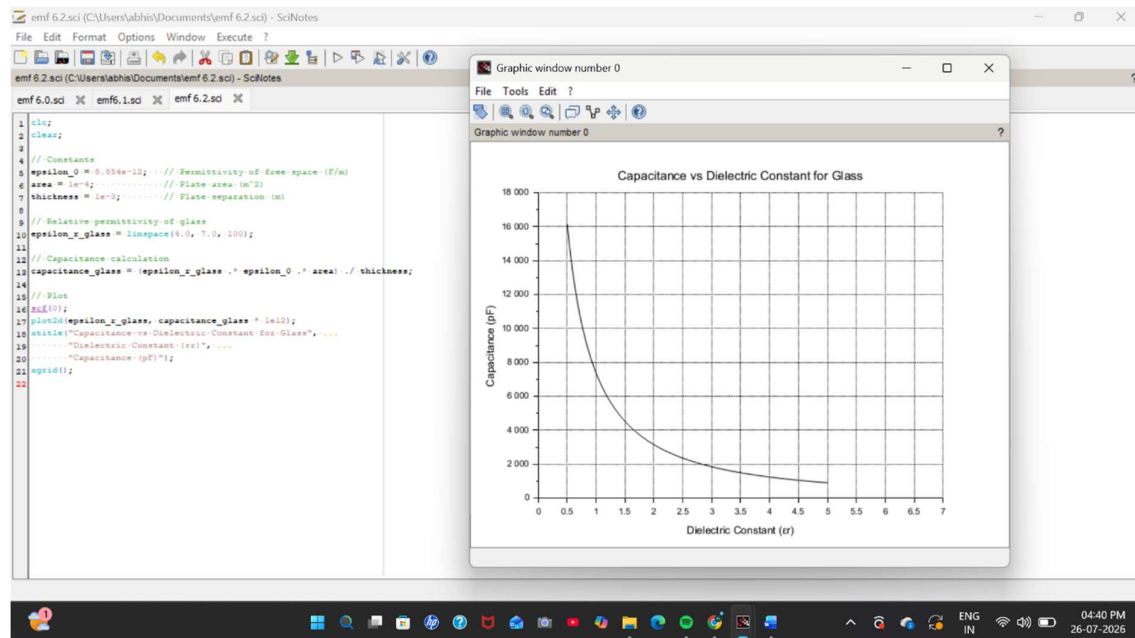
$$= \frac{5.3124 \times 10^{-11}}{0.03833}$$

$$C = 1.386 \times 10^{-9} \text{ F}$$

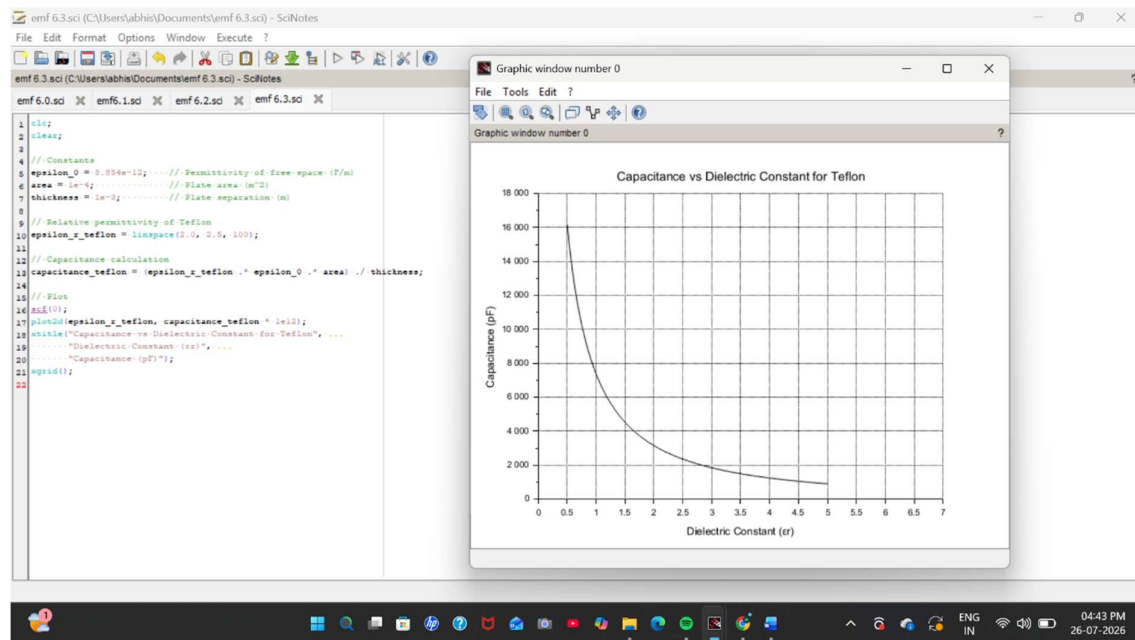
Case 1. AIR



Case 2. Glass:



Case 3. Teflon:



Result :

Thus the program was verified for Electric potential difference between two points in free space medium

using SCILAB was successfully.